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KEYED MEDICATION CARTRIDGE FOR USE IN A HANDHELD ORAL MEDICATION DISPENSING DEVICE

Abstract

The present disclosure provides a keyed medication cartridge for use in a handheld oral medication dispensing device. The medication cartridge is designed with alignment features for connecting to the medication dispensing device. The cartridge includes a shell and medication vial therein. In use, alignment features on the dispensing device pass through the shell and lock against the vial, allowing a user to depress the shell and activate a pump assembly to force medication from the medication cartridge.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This is a Continuation application of U.S. Ser. No. 18/601,329, filed Mar. 11, 2024, the entirety of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

(1) Field of Invention

[0002] The present invention relates to a medication cartridge and, more particularly, to a medication cartridge that is keyed to work with a handheld oral medication dispensing device.

(2) Description of Related Art

[0003] The field of medication treatment has grown increasingly complex in recent years, with patients frequently needing to manage and administer multiple types of medication. Under such circumstances, it is vital to provide reliable, accessible, and simple methodologies to patients for dispensing their medications. Many existing dispensing devices, however, tend to be cumbersome, confusing, or inefficient. Often, they rely on patients having to manually change their medication reminders or products, which can involve complicated steps or create potential for mistakes, like grabbing the wrong medication or dispensing an incorrect dosage. Furthermore, mistakes may potentially have serious health implications. Hence, it's vital to design an easy-to-use and fail-safe medication dispensing device.

[0004] Further, existing oral medications are typically taken as tablets or syrups. Such medication delivery items may be functional, but often take time for efficacy due to digestive and other body related functions. Additionally, such medication delivery techniques are inconvenient to carry on ones person or inconspicuously use. Thus, a need exists for a unique oral medication dispensing device and keyed medication cartridge that provides an alternative to tablet and syrup-type medications.

SUMMARY OF INVENTION

[0005] The present disclosure is directed to a medication cartridge that is keyed to work with a handheld oral medication dispensing device. The purpose of this design is not only to make medication administration simpler and more efficient but also to mitigate potential errors associated with manual cartridge switching. The dispensing device provides user authentication and employs a simple yet effective actuation mechanism to operate the medication cartridge, further enhancing safety measures. The innovative design of the keyed connector and the configuration of the cartridges can significantly aid patients manage and administer their medication effectively and safely.

[0006] In one aspect, the medication cartridge is an elongated cartridge with one or more alignment channels that are formed to align with and receive one or more alignment features of an oral medication dispensing device. The elongated cartridge includes a shell having a dispensing aperture, and a vial slidably positioned within the shell, the vial having a pump assembly with an outlet aligned with the dispensing aperture.

[0007] In another aspect, the one or more alignment channels includes a pair of opposing channels

formed along the vial.

[0008] In another aspect, the shell includes a pair of opposing slots that align with the pair of opposing channels.

[0009] Further, the vial is slidable within the shell between an expanded state and a compressed state, such that when transitioning to the compressed state, the pump assembly is actuated to pump medicine from the outlet and through the dispensing aperture.

[0010] In another aspect, the medication cartridge has a bottom end and a top end, such that each of the pair of opposing slots has a first slot that rises upward from the bottom end toward the top end to a first junction, and then a second slot that traverses downward at an angle from the first junction toward a second junction, and then a third slot that rises upward again from the second junction toward the top end.

[0011] In yet another aspect, each of the pair of opposing channels rises upward from the bottom end toward the top end to the first junction, and then traverses downward at an angle from the first junction toward the second junction, where each of the pair of opposing channels terminates in a channel terminal.

[0012] Further, the medication cartridge is formed such that depression of the top end of the medication cartridge forces the shell downward while the one or more alignment features pass through the third slots while remaining affixed within the channel terminals, thereby forcibly sliding the vial within the shell from the expanded state to the compressed state and actuating the pump assembly.

[0013] Finally, as can be appreciated by one in the art, the present invention also comprises a method for forming and using the invention as described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The objects, features and advantages of the present invention will be apparent from the following detailed descriptions of the various aspects of the invention in conjunction with reference to the following drawings, where:

[0015] FIG. 1 shows an exemplary illustration a handheld medication dispensing device and medication cartridge according to various embodiments of the present invention;

[0016] FIG. 2A shows an interior-view of the medication dispensing device, depicting a cartridge carrier;

[0017] FIG. 2B shows an interior-view of the medication dispensing device, depicting the medication cartridge inserted into the cartridge carrier;

[0018] FIG. 3 is an interior-view illustration of the medication dispensing device, depicting an activator mechanism as operably connected with the cartridge carrier;

[0019] FIG. 4 shows an exemplary illustration of the cartridge carrier and medication cartridge;

[0020] FIG. 5A shows a side-view illustration of the medication cartridge;

[0021] FIG. 5B shows a top-view illustration of the medication cartridge;

[0022] FIG. 5C is a cross-sectional, side-view illustration of the medication cartridge, taken from line A-A of FIG. 5B;

[0023] FIG. 6 shows an exploded-view illustration of the medication cartridge;

[0024] FIG. 7A shows a front-view illustration of the medication cartridge;

[0025] FIG. 7B shows a right, side-view illustration of the medication cartridge;

[0026] FIG. 7C shows a left, side-view illustration of the medication cartridge;

[0027] FIG. 7D shows a rear-view illustration of the medication cartridge;

[0028] FIG. 7E shows a top-view illustration of the medication cartridge;

[0029] FIG. 7F is a cross-sectional, side-view illustration of the medication cartridge, taken from

line A-A of FIG. 7C;

[0030] FIG. 7G is an illustration of a lotion pump assembly of the prior art;

[0031] FIG. 7H is an illustration of an example pump assembly as applied to the medication cartridge in accordance with various embodiments of the present invention;

[0032] FIG. 8A shows an exemplary illustration of the cartridge carrier and medication cartridge;

[0033] FIG. 8B shows an exemplary illustration of the cartridge carrier and medication cartridge, depicting the medication cartridge as being inserted into the cartridge carrier;

[0034] FIG. 9 shows an exemplary illustration of the medication cartridge inserted into the cartridge carrier;

[0035] FIG. 10A shows an exemplary illustration of the medication cartridge inserted into the cartridge carrier, depicting the medication cartridge in a stored position;

[0036] FIG. 10B shows an exemplary illustration of the medication cartridge inserted into the cartridge carrier, depicting the cartridge carrier rotating to move the medication cartridge up and toward the dispensing position;

[0037] FIG. 10C shows an exemplary illustration of the medication cartridge inserted into the cartridge carrier, depicting the cartridge carrier as rotated to move the medication cartridge up and into the dispensing position;

[0038] FIG. 11A shows an exemplary illustration of the medication cartridge, depicting the vial sliding within the shell;

[0039] FIG. 11B shows close-up view of Detail A, taken from FIG. 11A;

[0040] FIG. 11C shows an exemplary illustration of the medication cartridge, depicting the vial within the shell;

[0041] FIG. 12A shows an exemplary illustration of the medication cartridge within the cartridge carrier, depicting the medication cartridge in the dispensing position and being pressed downward to activate the pump assembly and dispense medication from the medication cartridge; and

[0042] FIG. 12B shows an exemplary illustration of the medication cartridge within the cartridge carrier, depicting the medication cartridge in the dispensing position and being pressed downward to activate the pump assembly and dispense medication from the medication cartridge.

DETAILED DESCRIPTION

[0043] The present invention relates to a medication cartridge and, more particularly, to a medication cartridge that is keyed to work with a handheld oral medication dispensing device. The following description is presented to enable one of ordinary skill in the art to make and use the invention and to incorporate it in the context of particular applications. Various modifications, as well as a variety of uses in different applications will be readily apparent to those skilled in the art, and the general principles defined herein may be applied to a wide range of embodiments. Thus, the present invention is not intended to be limited to the embodiments presented, but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

[0044] In the following detailed description, numerous specific details are set forth in order to provide a more thorough understanding of the present invention. However, it will be apparent to one skilled in the art that the present invention may be practiced without necessarily being limited to these specific details. In other instances, well-known structures and devices are shown in block diagram form, rather than in detail, in order to avoid obscuring the present invention.

[0045] The reader's attention is directed to all papers and documents which are filed concurrently with this specification and which are open to public inspection with this specification, and the contents of all such papers and documents are incorporated herein by reference. All the features disclosed in this specification, (including any accompanying claims, abstract, and drawings) may be replaced by alternative features serving the same, equivalent or similar purpose, unless expressly stated otherwise. Thus, unless expressly stated otherwise, each feature disclosed is only one example of a generic series of equivalent or similar features.

[0046] Furthermore, any element in a claim that does not explicitly state "means for" performing a

specified function, or “step for” performing a specific function, is not to be interpreted as a “means” or “step” clause as specified in 35 U.S.C. Section 112, Paragraph 6. In particular, the use of “step of” or “act of” in the claims herein is not intended to invoke the provisions of 35 U.S.C. 112, Paragraph 6.

[0047] Please note, if used, the labels left, right, front, back, top, bottom, forward, reverse, clockwise and counter clockwise have been used for convenience purposes only and are not intended to imply any particular fixed direction. Instead, they are used to reflect relative locations and/or directions between various portions of an object.

(1) Description

[0048] As noted above, the present disclosure is directed to a medication cartridge that is keyed to work with a handheld oral medication dispensing device. As shown in FIG. 1, the handheld medication dispensing device **100** is designed primarily to securely store and dispense medication from a medication cartridge **102** of the present disclosure. The device **100** comprises a housing **104**, principally structured to protect the critical internal components while offering ergonomic suitability for handheld operation. The housing **104** exterior is designed with both convenience and comfort in mind, allowing a user to hold the device **100** and access the medication with ease. In an exemplary embodiment, the housing **104** may be configured in a variety of shapes, sizes, and materials to accommodate different user preferences and medication types.

[0049] In one embodiment, the device **100** includes an authentication mechanism **106** that is designed to authenticate a user's identification in various manners. The authentication mechanism **106** is any suitable mechanism, device, system, etc., that allows for user authentication, non-limiting examples of which include biometrics (such as fingerprint reader or facial recognition via a camera and app, etc.), a password or passcode system, a security token, or any other known authentication method or device. The authentication mechanism **106** is set up to permit the authenticated or designated user access to the operation of the said device **100**, thereby ensuring a highly secure and personal way of dispensing medication. In one embodiment and as illustrated, the authentication mechanism **106** includes a fingerprint reader and all associated components as necessary to allow for authentication with such a reader, including an integrated circuit, power (battery), etc. For example, in a first use, the authentication mechanism **106** can be configured to recognize the user and thereafter only allow access to the recognized user. Such a setup can be configured in any suitable manner as understood by those skilled in the art. As non-limiting example, a first use may require that a user place a finger on the authentication mechanism **106** which turns on the device **100** and provides indicia (e.g., via a light, digital screen, etc.) to notify the user to place their finger on the fingerprint reader. The fingerprint reader then reads and stores the fingerprint of the designated user. Thereafter, the activator mechanism can only be activated by the designated user upon placement of the finger on the fingerprint reader. In another example, software or a phone application can be used to wirelessly communicate (e.g., via Wi-Fi, Bluetooth, etc.) with the device **100** to program the device to only recognize the designated user via the authentication mechanism. In this aspect, the device **100** includes all necessary components to allow it to wirelessly communicate with an external device, non-limiting examples of which include Wi-Fi transceivers/components, Bluetooth components/gear, etc. In an alternative embodiment, the device **100** can simply be turned off and on (without an authentication mechanism), and includes a button or other similar feature to cause the device **100** to operate and actuate the activation mechanism as described in further detail below.

[0050] In one aspect, the authentication mechanism **106** is a biometric fingerprint reader linked to a mobile app, which can be used to prevent unauthorized medication dispensing, requiring user authorization via the app for initialization, and connecting to caregivers and personal networks to ensure medication adherence. The design of the device **100**, accessories, and app is centered around the need for an easier and better way to receive the right dose, at the right time, in the right way and frequency. In one aspect, controlled via the app, the handheld device **100** can alert users to their

medication schedule, while also notifying caregivers and family members about adherence. For restricted medications, the device **100** and associated app can restrict dosing until authorized by the prescribing physician. Tampering with the medication cartridge **102** triggers alerts to caregivers, pharmacists, and physicians, ensuring safe usage, especially for restricted medications by alerting through the app that the medication has not been taken as prescribed. In another aspect, the device **100** and medication cartridge **102** are formed to include any necessary components that allow the device **100** to recognize a specific medication cartridge **102**. As a non-limiting example, each medication cartridge **102** can be formed to include a Radio Frequency Identification (RFID) tag (or similar technology), while the device **100** includes an RFID reader (or other similar technology).

[0051] As shown in the interior views of FIGS. 2A and 2B, integrated within the housing **104** is a cartridge carrier **200**. The cartridge carrier **200** functions as a receptacle for the medication cartridge **102**. The cartridge carrier **200** includes a keyed connector, most importantly, innovatively designed to align with a specified type of medication cartridge **102**. This keyed connector serves as an interlocking structure that carefully guides the positioning and orientation of the cartridge **102**, allowing only for the insertion of a compatible medication cartridge **102**. The keyed connector along with the cartridge carrier **200** ensure the correct placement and alignment of the medication cartridge **102**, which can help eliminate potential misuse or cartridge damage, and fostering a smooth transition between stored and dispensing positions (as described in further detail below).

[0052] As shown in the interior view of FIG. 3, an activator mechanism **300** forms part of the configuration and is integrated within the housing **104**. The activator mechanism **300**, when engaged, prompts the cartridge carrier **200** to actuate. The involvement of the activator mechanism **300** ensures a controlled movement of the inserted medication cartridge **102** between two positions—a stored position and a dispensing position. Note that FIG. 2A depicts the medication cartridge in a stored position, while FIG. 3 depicts the medication cartridge in a dispensing position.

[0053] In the stored position, as shown in FIG. 2A, the medication is kept secure within the device **100**. The position ensures safety by preventing unintentional discharge of medication when not in use. When the activator mechanism **300** is activated, the cartridge carrier **200** rotates to lift up the medication cartridge **102** into the dispensing position. When this dispensing position is selected, the medication cartridge **102** is correctly oriented to dispense the medication through a dispensing aperture (described in further detail below) integrated into the medication cartridge **102**.

[0054] The link between the activator mechanism **300** and the cartridge carrier **200** allows for reliable and essential movement control. This means the user can confidently operate the device **100** knowing the medication is dispensed accurately and safely.

[0055] The cartridge carrier **200** is formed in any suitable manner to allow for activation by the activator mechanism **300**, which results in transitioning the medication cartridge **102** between the stored and dispensing positions. In an embodiment and as shown in FIG. 3, the cartridge carrier **200** is rotatable within the housing **104**. The cartridge carrier **200** may be formed of materials that possess strength and durability such as, but not limited to, metals, polymers, and/or composites. The size, shape, and dimensions of the cartridge carrier may vary as per the application requirements.

[0056] The housing **104** may be constructed to accommodate the rotatable cartridge carrier **200**. The construction of the housing **104** can also encompass different types of materials that provide the necessary strength and durability inclusive of, but not limited to metals, polymers, composites, among others. The interior of the housing **104** is ideally designed to allow the cartridge carrier **200** to rotate through actuation of the activator mechanism **300**.

[0057] The activator mechanism **300**, ensuring the cartridge carrier's **200** rotation within the housing **104**, can be implemented through a variety of means, such as mechanical gears, electric motors, a battery, manually operated mechanisms, integrated circuits, or any combination thereof. This activator mechanism **300** administration permits the user to precisely position the cartridge carrier **200** within the housing **104** between a first position (in which the medication cartridge **102**

is inserted into the cartridge carrier **200** and is stored in a stored position) and a second position (in which the medication cartridge **102** is lifted from the housing **104** to a dispensing position). The activator mechanism **300** can then be activated to rotate the cartridge carrier **200** in an opposite direction to return the medication cartridge **102** to the stored position.

[0058] In one aspect, the dispensing device **100** includes a locking mechanism that is configured to selectively lock/unlock the medication cartridge **102** within the cartridge carrier **200**. The locking mechanism may involve an interlocking system, a magnetic attachment, or any other secure attachment mechanism. As a non-limiting example, the locking mechanism may be a spring-loaded detent style 'click-in' and 'click-out' mechanism formed at the bottom of the medication cartridge **102** and cartridge carrier **200**. In another aspect, the locking mechanism can be an electronically actuated lock (e.g., mechanized tab, etc.) that locks the medication cartridge **102** into the cartridge carrier **200**. For example, the exterior surface of the medication cartridge can include a slot that is formed to accommodate a tab that is electronically moved (e.g., turned/slid, etc.) into the slot when locking and, in the alternative, pulled from the slot when released. As yet another non-limiting example, the locking mechanism can include programming such that after the cartridge **102** is inserted into the cartridge carrier **202**, the cartridge carrier **200** is rotate partially until alignment features (described in further detail below and depicted as element **406**) are positioned in the middle of the second slot (described in further detail below and depicted as element **710**), with the cartridge carrier **200** then stopped until it is desired to rotate further to position the medication cartridge **102** in the dispensing position. With the alignment features in the middle of the second slot, the cartridge **102** is effectively locked within the cartridge carrier **200**. Thus, as can be appreciated by those skilled in the art, there are several mechanisms that can be employed to selectively lock the medication cartridge **102** into the cartridge carrier **200**.

[0059] In another aspect, the inclusion of the locking mechanism may also be included to secure the cartridge carrier **200** at any desired rotational position, thus preventing unintentional and unwanted movement when the cartridge carrier **200** is set in a particular position. This locking mechanism can be integrated into the cartridge carrier **200**, the housing **104**, the activator mechanism **300**, or combinations of these components. This rotatable cartridge carrier **200** system installed within a housing **104** assembly may have wide ranging applications, such as, for instance, accommodating various cartridges for providing users with quick and easy access to different medications and dosages, multiple configurations, interchangeability of components, and more. Thus, the present one embodiment discloses a rotatable cartridge carrier **200** system, promoting ease of use, enhancing performance, providing efficient operation, flexibility, improved control and precision, leading to a more efficient and user-friendly functionality.

[0060] In one embodiment and as shown in FIG. 4, the cartridge carrier **200** includes both a proximal end **400** and a distal end **402**. The proximal end **400** of this unit is open, and it is specifically structured to receive the medication cartridge **102**. This can be any kind of medication cartridge **102**, and the open design simplifies the process of interchanging cartridges of differing medications, volumes, or other specifications, thus making it broadly applicable and versatile.

[0061] The distal end **402** of the cartridge carrier **200** includes a key component of this one embodiment—a gear wheel **404**. This gear wheel **404** is not a mere inclusion but is operably connected to the activator mechanism **300**. The gear wheel **404** and activator mechanism **300** are conceptually designed to work in harmony. In operation, when the user deploys the activator mechanism **300**, the included gear wheel **404** is rotated, thereby rotating the cartridge carrier **200**.

[0062] The gear wheel **404** is a significant element in the functioning of this device **100** mainly due to its interaction with the activator mechanism **300**. The gear wheel **404**, upon the interaction with activator mechanism **300**, contributes to the precision with which the cartridge carrier **200** is rotated to position the medication cartridge **102** between the stored and dispensing positions. Its size, teeth count, and interaction with other gears if connected, contribute to its function.

[0063] Moreover, the position of the gear wheel **404** at the distal end **402** of the cartridge carrier

200 may also assist in a user-friendly, ergonomic design. The gear wheel's **404** location ensures a seamless connection between the mechanical parts of the activator mechanism **300** and the carrier cartridge **200**, augmenting the device's **100** overall performance.

[0064] As noted above and referring again to FIG. 3, the device **100** includes an activator mechanism **300** to selectively turn the gear wheel **404** and, by extension, rotate the cartridge carrier **200**. The activator mechanism **300** includes any components as necessary to rotate the gear wheel **404**. In one embodiment, the activator mechanism **300** comprises of at least a battery **304**, a motor **302**, and one or more gears **306** that are operably connected to the gear wheel **404**. The battery in the device **100** forms the power source, capable of providing a stable and reliable supply of power to the other components. The size, capacity, and type of the battery **304** may be customized according to design preferences, device requirements, or various other factors, while being compliant with standard safety and operation guidelines. In some aspects, the battery **304** is encased within the device **100** and intended for device **100** replacement upon depletion of the battery **304**, while in other aspects, it is replaceable. In yet other aspects, the battery **304** is rechargeable using a charging port **308** or any other charging means as known to those skilled in the art.

[0065] The motor **302** functions as the primary mechanical component that is responsible for movement or force in the device **100**. Upon activation, the motor **302** draws power from the battery and begins to turn to rotate any of the one or more gears and the operably connected gear wheel **404**. The type, size, and characteristics of the motor **302** can be adjusted based on the requirements of the device, anticipated load, desired performance factors among other parameters.

[0066] The one or more gears represent a transmission mechanism, operably connected to the motor **302** and the gear wheel **404**. Upon the activation of the device **100**, the rotational force produced by the motor **302** is transferred to these gears. The gears enable the transfer of power from the motor **302** to the gear wheel **404**. Upon receiving the force from the one or more gears of the activator mechanism **300**, the gear wheel **404** begins to turn. This turning of the gear wheel **404** then contributes to the rotation of a cartridge carrier **200**.

[0067] Referring again to FIG. 4, one embodiment pertains to an enhanced cartridge carrier **200** system, specifically designed with one or more alignment features **406** that protrude into the cartridge carrier **200** for a keyed connection and accurate cartridge **102** positioning, thereby improving the overall functionality of the cartridge **102** and the cartridge carrier **200** system.

[0068] The alignment features **406** of one embodiment are distinguishable protrusions present on the inner surface of the cartridge carrier **200**. These alignment features **406** serve to guide the cartridge **102** into the correct position within the cartridge carrier **200** when installing a cartridge **102** into the system. These protruding alignment features **406** also help to prevent misalignment and displacement of the cartridge **102** while it is installed and during operation. Notably and as described below in further detail, the alignment features **406** are critical in transitioning an inserted cartridge **102** between the stored and dispensing position.

[0069] The alignment features **406** are strategically located within the cartridge carrier **200** in a manner to engage with corresponding features or portions on the cartridge **102**. When the cartridge **102** is inserted into the carrier **200**, the protruding alignment features **406** guide it into the correct position as the features **406** fit into compatible sections (i.e., alignment channels **408**) of the cartridge **102**. This results in a snug and correct fit of the cartridge **102** within the cartridge carrier **200**.

[0070] Furthermore, these alignment features **406** may come in various shapes and sizes and can be constructed using various materials that are suitable for the system and the intended application. They can be rigid to withstand the pressure exerted by the cartridge **102** as it is inserted into the cartridge carrier **200**, or flexible to allow for slight variations in cartridge **102** size and shape.

[0071] For further understanding, FIGS. 5A through 5C depict side, top, and cross-sectional views, respectively, of the cartridge carrier **200** and alignment features **406**. These features **406**, protruding

into the interior of the cartridge carrier **200**, support the accurate placement of cartridges **102** within the device. Thus, the functionality and efficiency of the overall dispensing device **100** are significantly improved along with consistent cartridge **102** performance ensured by these alignment features **406**. This inventive cartridge carrier **200** system design genuinely adds value to applications where precise alignment of cartridges **102** in their carrier systems is crucial. This one embodiment indeed ensures an innovative update for improving the efficiency of cartridge insertion and alignment in cartridge carrier systems, potentially paving the way for advancements in devices utilizing such systems.

[0072] As noted above and referring again to FIG. **4**, the present disclosure also provides a unique medication cartridge **102** that can be inserted within a cartridge carrier, thus providing a novel means to carry, store and administer medication. The essence of this one embodiment lies in the exclusive design of the medication cartridge **102** and the cartridge carrier that is not only easy to operate but also provides secure storage and effortless administration of the medication.

[0073] The medication cartridge **102** is an integral part of the inventive concept and is designed to contain distinct types of medications, be it in liquid, gel, or cream, or other dispensable forms of medication. The cartridge **102** can be made up of pharmaceutically safe materials ensuring no chemical reactions occur when in contact with the medication. The cartridge **102**, itself can showcase cylindrical or other geometrical shapes as required by shape of the cartridge carrier **200**. The medication cartridge **102** and cartridge carrier **200** partnership provides an ideal solution in various medical environments, such as hospitals, pharmacies, nursing homes, or even domestic settings. It proves to be beneficial where multiple medications should be stored and administered safely and hygienically. It combines innovation with convenience, improving medication storage, and administration methodologies.

[0074] As noted above and as shown in FIG. **6**, the cartridge **102** itself is also designed to provide a marked improvement over other medication containers. It is envisioned to include markings or identification systems for easily recognizing the type of medication stored within each cartridge **102**. The real innovation lies in the incorporation of one or more channels **408** and aligned slots **606** within the cartridge **102**. These are constructed meticulously within the cartridge **102** and serve a distinct function of aligning and receiving one or more alignment features **406**. As will be evident below, these channels **408** and slots **606** engage with the aforementioned alignment features **406** to allow for controlled operation of the device **100**.

[0075] Further and as shown throughout FIGS. **6** through **7F**, the present disclosure provides an improved configuration for a medication cartridge **102**, particularly advantageous for medications that require enhanced storage for securely maintaining their efficacy and quality. The innovative design of the medication cartridge **102** includes a robust shell **600** that has a dispensing aperture **604** designed for the release of the medication stored within a medication vial **602**.

[0076] The shell **600**, formed from long-lasting and reliable material (e.g., plastic, metal, etc.), is shaped to house the vial **602** securely and efficiently. This shell **600** acts as a protective layer for the vial **602** placed inside, while maintaining the required conditions for the preservation of the medication. The dispensing aperture **604** deployed on the shell **600** is engineered in a way to align with the outlet **702** of the vial **602**, ensuring precise and mess-free dispensation of the medication.

[0077] Inside the shell **600** is a medication vial **602**, which is shaped and positioned in a manner that it can slide without difficulty within the shell **600**. This ability of the vial **602** to slide enables convenient load and unload of the medication. The vial **602** includes a pump assembly **700** that is equipped with an outlet **702**. The design and positioning of this outlet **702** are such that it aligns perfectly with the dispensing aperture **604** on the shell.

[0078] The pump assembly **700** is any suitable assembly that can be affixed with a container (i.e., the vial **602**) to allow for pumping or otherwise dispensing a medication or fluid therefrom, a non-limiting example of which includes a compression or spring-loaded pump, similar to the pump mechanism as used in a standard lotion bottle. For reference, FIG. **7G** provides an illustration of an

example pump assembly as provided for in the prior art, while FIG. 7H provides a non-limiting example of the pump assembly as modified and applied to the present invention. The pump assembly of the prior art (as shown in FIG. 7G) includes a clamp **1**, press head **2**, press lever **3**, coat **4**, screw cap **5**, connect cap **6**, gasket **7**, piston **8**, piston seat **9**, spring **10**, spring seat **11**, valve **12**, housing **13**, and a straw **14**. As applied to the present invention (as depicted in FIGS. 7F and 7H), the pump assembly **700** is connected to the vial **602**, while the shell **602** serves as the press head that is used to press down the piston **8** to force the fluid through the straw **14** and out of the dispensing aperture **604**. In some instances the pump assembly **700** can be screwed onto the vial **602**; however, it should be understood that the invention is not intended to be limited thereto as the pump assembly **700** can be attached in any suitable manner between the vial **602** and shell **600** to allow for dispensing fluid from the vial **602** and out of the dispensing aperture **604**. Further, it should be understood that the specific pump assembly **700** as depicted in FIG. 7H is provided as a non-limiting example of a suitable pump assembly and that the invention is not intended to be limited thereto as other mechanisms or assemblies can be used to dispense the fluid upon actuation. For example, the pump assembly can be formed as an airless pump, a peristaltic pump, a diaphragm pump, a pressurized canister style pump, or any other configuration in which activation of the assembly causes the fluid to dispense.

[0079] For illustrative purposes in comparison to a lotion bottle, the vial **602** would serve as the container to hold the fluid, while the shell **600** serves as the pump head that can be depressed to pump the fluid, with the various components of the pump assembly **700** positioned therebetween. In operation and when the medication cartridge **102** is moved to the dispensing position, the alignment features **406** operate to hold the vial **602** in place (due to their locking position in the channel terminal as described in further detail below), while a user can selective press the shell **600** downward around the vial **602** to activate the pump assembly **700** and force fluid from the vial **602**.

[0080] For example, the vial **602** includes a reservoir **720** in which the medication is stored. The medication can be in a gas or fluid form; however, desirably and in one aspect, the medication is fluid and can be squirted from the medication cartridge **102** (and vial **602**) via actuation of the pump assembly **700**. As apparent from the description further below, the pump assembly **700** can be easily operated to cause the medication to be dispensed from the vial **602** through its outlet **702**. When a force is applied on the pump assembly **700**, it triggers the release of the medication inside of the vial **602**, causing the medication to travel from the vial **602**, passing through the aligned outlet **702** and dispensing aperture **604**, where it is released to the user.

[0081] As noted above, the pump assembly **700** is any suitable assembly, mechanism, or device that can be affixed with the vial **602** to allow for selective dispensing therefrom. As can be appreciated by those skilled in the art, a non-limiting example of such a pump assembly **700** is a spring-loaded system utilizing a stacked one-way valve system to temporarily create a vacuum, allowing fluid to be aspirated from the reservoir **720** and dispensed through the outlet **702** and dispensing aperture **604** upon depression of the shell **600** around the vial **602**.

[0082] As noted above, the medication cartridge **102** is generally formed of two components, a shell **600** and a medication vial **602**. The vial **602** is sized to be secured within and slide within the shell **600**. As shown, the medication cartridge **102** includes a top end **704** and a bottom end **706**. Although the shell **600** in FIG. 6 is illustrated as having a cap component **608** at the top end **704**, the invention is not intended to be limited thereto as the shell **600** can be formed such that the cap component **608** is integrally formed with the shell **600** as a single unit. In such an aspect, a hole (e.g., sealable hole) can be formed at the bottom end **706** (or top end **704**) of the shell **600** to allow for insertion of the vial **602** into the shell **600**.

[0083] Notably and of particular importance, the vial **602** includes channels **408** that are aligned with the slots **606** on the shell **600**. These channels **408** and slots **606** are built in a manner to accurately align with the alignment features **406** of the cartridge carrier **200**. The number and size of these channels **408** and slots **606** can vary depending on the number of alignment features **406**

intended to be placed therein. These channels are purposefully built to receive these alignment features, allowing secure and precise positioning. This design ensures that the alignment features do not shift their placement, further guaranteeing the stabilization of the medication within the cartridge.

[0084] For example, one example embodiment includes a structure involving a pair of opposing channels **408** are formed on opposite sides of the vial **602**. These channels **408** are typically parallel to each other, though variations can occur. In one aspect, each of the pair of opposing channels **408** originates from the lowermost end or bottom end **706** of the vial **602**. The pair of opposing channels **408** rise upward from this point, extending towards the top end **704** of the vial **602**. This ascension continues until they reach a point known as the first junction **714**. It must be clarified that this first junction **714** is not an arbitrary point but is a strategically designed and located structural feature within the device to precisely position the cartridge **102** between stored and dispensing positions.

[0085] Once these channels **408** have reached the first junction **714**, the trajectory shifts. At this point, the channels **408** do not continue with the upward gradient. Instead, they commence a downward traversal toward the bottom end **706**. They proceed downwards, but not vertically downwards. This descent is carried out on an angle around the vial **602**, intended to enhance function and performance.

[0086] These angled downward channels **408** continue to move in the said direction until they reach an area deemed as the second junction **716**. It is at this point that these channels **408** cease to continue. They terminate at this juncture, each channel **408** ending in what is referred to as a channel terminal **718**. As will be apparent below, this distinctive configuration of the channels **708** enables them to perform their intended purpose more efficiently. It is important to consider the precise and specific architectural design of these channels **408**, including their starting point at the bottom end **706** of the vial **602**, their rise towards the first junction **714**, their angled descent towards the second junction **716**, which with respect to the channels, is the final termination at the channel terminal **718**. These particular design elements reveal a thoughtful consideration of their role within the device and contribute both independently and collectively to the overall functionality and performance of the device or system in question.

[0087] The present disclosure also provides an embodiment which is characterized by a pair of opposing slots **606** in the shell **600** that align with the pair of opposing channels **408**. More specifically, one embodiment is the configuration and interaction of these parts, enabling the pair of opposing protrusions **406** to pass through the pair of opposing slots **606** and into the pair of opposing channels **408** to allow for moving the cartridge **102** between the stored and dispensing positioned and, ultimately, sliding the vial **602** within the shell **600** for dispensation of the medication.

[0088] The pair of opposing slots **606** are formed through the surface of the shell **600** to allow access to the channels **408** therein. These slots **606** are designed with precise measurements to fit and accommodate corresponding elements, namely, a distinct pair of opposing protrusions **406**. The opposing slots **606** are carved or formed into the shell **600** and aligned in such a way that they maintain a clear, unobstructed path for the protrusions **406** to pass through them with ease. The unique arrangement allows for the insertion of the pair of opposing protrusions **406**, which are another critical characteristic of this one embodiment. These protrusions **406**, with their matching dimensions to the slots **606**, are designed to glide smoothly through the opposing slots **606** and into the aligned opposing channels **408**. The path provided by the slots **606** guides and directs the protrusions **406** into the appropriate channels **408**. Consequently, this ensures a specific interaction between the three parts: the opposing protrusions **406** smoothly pass through the opposing slots **606**, and they are then guided into the opposing channels **408**, maintaining a suitable and secure connection.

[0089] Further, the medication cartridge **102** presents an innovative design with respect to the pair

of opposing slots **606**. Each of the pair of opposing slots **606** is formed of a first slot **708**, a second slot **710**, and a third slot **712**. These slots are strategically positioned and oriented for the optimal functioning of the device. The first slot **708** in each pair begins at the bottom end **706** of the shell **600** and extends upward toward the top end **704**. It is arranged in such a way that it mandates a route which ascends from the base to a designated position along the length of the cartridge **102**, otherwise referred to as the first junction **714**.

[0090] Subsequent to the first junction **714**, the structure of the shell **600** features a second slot **710** for each pair. Instead of following the upward trajectory of its predecessor, this second slot **710** takes on a unique characteristic of its own by providing an angled path that descends from the first junction **714**. This downward movement proceeds until it reaches a designated point along the body of the shell **600**, referred to herein as the second junction **716**. Moreover, the slot design of the shell **600** reverts back to an upward direction in the subsequent and final arrangement of each pair-the third slot **712**. Commencing from the second junction **716**, this slot ascends once again (e.g., vertically) towards the top end **704** of the cartridge. The distinct positions and paths of the slots **708**, **710**, and **712** provide a sophisticated layout that enhances the fundamental operations of this medication cartridge **102**. Notably, while the channel **408** terminates at the second junction **716** or channel terminal **718**, the third slot **712** proceeds upwards from the second junction **716**. This differentiation allows for the selective dispensing of the medication therein. This innovative structure enhances the controlled dispensing of medication, making this cartridge **102** particularly advantageous in the administration of drugs. The angles and special slot layout improves the overall efficiency and operational effectiveness of the delivery system, which in turn can broaden its potential applications in the medical and pharmaceutical fields. The precise orientation of the slots from bottom to top and their alternating directions contribute to creating a balanced and smooth transition of the medication within the cartridge **102**. The level of control provided by this inventive structure is a significant evolution in medication delivery system designs.

[0091] For further understanding, FIGS. **8A** and **8B** depict a medication cartridge **102** as being installed or otherwise inserted into the cartridge carrier **200**. As shown, the alignment features **406** (i.e., opposing protrusions) provided a keyed connection and alignment by passing through the channels **408** and slots **606** to allow the cartridge **102** to be contained within the carrier **200**. As shown in FIG. **9**, the alignment feature **406** has travelled through the first slot **708** and channel **408** to rest at the first junction **714**. In this position, the medication cartridge **102** is held within the dispensing device in a stored position. Since the alignment feature **406** is at a fixed location within the cartridge carrier **200**, and because the second slot **710** and channel **408** descend downwards at an angle from the first junction **714**, rotation of the cartridge carrier **200** necessarily forces the medication cartridge **102** upwards within the cartridge carrier **200**. These aspects are further depicted in FIGS. **10A** through **10C**.

[0092] FIG. **10A** depicts the medication cartridge in a stored position. As shown, the alignment feature **416** rests within the first junction **714**. As shown in FIG. **10B**, rotation **1000** (via the gear wheel **404**) of the cartridge carrier **200** forces the alignment feature **416** to traverse through the second slot **710** and channel **408** away from the first junction **714**. In doing so and as shown, the medication cartridge **102** is lifted upwards within the cartridge carrier **200**. Further rotation **1000** of the cartridge carrier **200**, as shown in FIG. **10C**, results in the alignment feature **416** to traverse until it rests within the second junction **716** and the channel **408** terminates at the channel terminal **718**. Notably, when the alignment feature **416** is at the second junction **716**, the medication cartridge **102** is lifted from the dispensing device **100** into the dispensing position. Thus, the journey of the protrusions commences from the first junction **714**, progressing towards the second junction **716**. As a consequence of this carefully guided pathway, the medication cartridge **102** experiences an upward shift. This movement is instrumental in the transition of the medication cartridge **102** from what is denoted as the 'stored position' to its final location, the 'dispensing position'.

[0093] While the cartridge **102** is in the dispensing position, the vial **602** therein is in the expanded state and ready to be compressed to dispense medication. As shown in FIGS. **11A** through **11C**, although the channel **408** in the vial **602** has terminated, the slot in the shell **600** continues upwards as the third slot **712**. Thus, when the medication cartridge **102** is in the dispensing position, a user can selectively press down on the medication cartridge **102** to cause the vial **602** to slide within the shell **600**. As shown in FIG. **11C**, due to the stepped inner wall **1100** and end of the channel **408**, the vial **602** is held in place by the alignment features **406** and only the outer shell **600** can travel downward when pressed down, causing the shell **600** to slide downward over the vial **602** and allowing the pump assembly to be actuated and spray/dispense the medication. This feature is further illustrated in FIGS. **12A** and **12B**. As shown, pressing downward **1200** on the medication cartridge **102** causes the alignment features **406** to travel upwards through the third slot **712**, thereby forcing the shell **600** downward while holding the vial **602** in place. While the shell **600** is forced downward, it compresses the vial **602** from the expanded to compressed state. The transition from the expanded to compressed state activates the pump assembly to force medication **1202** from the vial **602** and through the dispensing aperture **604**, similar to a squirt of medication. The pump assembly can be configured as a spring-loaded pump mechanism that lifts the shell **600** after being compressed, thereby returning the alignment features **406** to the second junction **716**. Once dispensed, the dispensing device **100** and associated activator mechanism **300** can be activated to turn the gear wheel **404** in an opposite direction as previously rotated. Such a rotation lowers the medication cartridge **102** back to the stored position as the alignment features **406** are forced back to the first junction **714** (shown in FIG. **10A**).

[0094] In one aspect, the device **100** is formed to sense once a predetermined number of squirts (e.g., one, etc., as prescribed) have been initiated so that it automatically activates the activator mechanism **300** to return the medication cartridge **102** back to the stored position, thereby retracting the medication cartridge **102** and inaccessibility after user. For example, magnetic sensors, a trigger, electrical sensors, light sensors, or any other means for determining if medication has been dispensed from the vial **602** can be included within the device **100**. As a non-limiting example, a magnet can be positioned at the bottom of the shell **600** while a magnetic reed switch is similarly positioned at the bottom of the cartridge carrier **200**. Pressing the medication cartridge **102** and its shell **600** downward within the cartridge carrier **200** brings the magnet into close proximity of the magnetic reed switch to notify the associated electronics (integrated circuit, etc.) that a single squirt has been dispensed. After the predetermined number of squirts, the activator mechanism is activated to rotate the cartridge carrier **200** and return the medication cartridge **102** to the stored position to prevent further and unauthorized use. In another aspect, the device **100** can include any necessary components (integrated circuit, etc.) to allow for a timed configuration in which the medication cartridge **102** is held in the dispensing position. For example, the device **100** can be programmed such that after authentication and movement of the medication cartridge **102** in the dispensing position, it is returned to the stored position after a predetermined or preprogrammed amount of time (e.g. 2 seconds, etc.).

[0095] As noted above, a unique design feature allows for the vial **602** to exist in two distinct states; namely an expanded state and a compressed state. The transition between these two stages is critical to the functionality of the device. When it moves from the expanded state towards the compressed state, the incorporated pump assembly **700** is triggered into action. This incorporation of a pump assembly **700** and its associated operation indicates an upbeat sophistication of the one embodiment, going beyond traditional static medical dispensing devices. The purpose of this actuation is specifically to ensure the controlled release of the medicine within the vial **602**. On the actuation of the pump assembly **700**, the medicine **1202** is pushed out of the vial's **602** outlet. This outflow proceeds under the controlling and moderating role of the pump assembly **700**, guaranteeing a controlled and managed release of the medicine.

[0096] In summary, the present disclosure provides an innovative medication cartridge **102** for use

with a dispensing device **100**. The core of this medication dispensing device **100** revolves around a unique cartridge carrier **200** design, integrated with a pair of opposing protrusions **406** and a uniquely designed medication cartridge that is functionally designed to operate based on the positioning of the protrusions **406** after user authentication. As noted above, in one aspect, the device **100** includes an authentication mechanism **106**. Upon successful authentication, the authentication mechanism **106** triggers the activator mechanism **300**. In this context, the activation entails the initiation of moving the medication cartridge **102**. When activated, it can efficaciously induce the cartridge carrier **200** mechanism to actuate, progressing the inserted medication cartridge **102** from a stored position to a dispensing position. Maximal effectiveness is ensured as the activator mechanism **106** is engendered only after a successful user authentication process, thereby verifying that the medication is dispensed to the appropriate user. This sequence not only ensures secure and controlled dispensing of the medication but also minimizes errors and the potential for unlawful access. In one aspect, the interconnectedness between the authentication mechanism **106** and the activator mechanism **300** forms the crux of this system, thereby augmenting the overall safety and operation of the dispensing device **100**. Moreover, this intricate mechanism, in coordination with the user and the cartridge carrier **200**, ensures that the process of moving the medication cartridge **102** from a stored position to a dispensing position is not only smooth but also efficient and secure.

[0097] Thus, in one aspect, a purpose of this inventive operation is to modify the positioning of the medication cartridge **102** from a state of storage to a dispensing state, thereby facilitating the delivery of medication in a controlled, efficient, and timely manner. The operational process involved in use of the dispensing device **100** advances through several stages, starting with the activation of the activator mechanism **300**. The initiation of this mechanism **300** leads to a rotational motion in the cartridge carrier **200**, a critical movement that fosters the maneuvering of the pair of opposing protrusions **406** (i.e., alignment features). This rotation is precisely designed to stimulate this movement, making the entire process efficient and seamless. The pair of opposing protrusions **406** are guided in their motion by a pair of opposing slots **606** in conjunction with a pair of opposing channels **408**. These channels **408** and slots **606** function in harmony, not only providing a safe passage but also controlling the direction of the movement of these protrusions **406** while guiding the medication cartridge **102** between a stored position and a raised dispensing position. The unique features of the slots **606** and channels **408** also allow a user to compress the vial **602** within the medication cartridge **102** to dispense medication **1202**.

[0098] Finally, while this invention has been described in terms of several embodiments, one of ordinary skill in the art will readily recognize that the invention may have other applications in other environments. It should be noted that many embodiments and implementations are possible. Further, the following claims are in no way intended to limit the scope of the present invention to the specific embodiments described above. In addition, any recitation of “means for” is intended to evoke a means-plus-function reading of an element and a claim, whereas, any elements that do not specifically use the recitation “means for”, are not intended to be read as means-plus-function elements, even if the claim otherwise includes the word “means”. Further, while particular method steps have been recited in a particular order, the method steps may occur in any desired order and fall within the scope of the present invention.

Claims

1. A medication cartridge, comprising: an elongated cartridge with one or more alignment channels that are formed to align with and receive one or more alignment features of an oral medication dispensing device.
2. The medication cartridge as set forth in claim 1, wherein the elongated cartridge includes: a shell having a dispensing aperture; a vial slidably positioned within the shell, the vial having a pump

assembly with an outlet aligned with the dispensing aperture.

3. The medication cartridge as set forth in claim 2, wherein the one or more alignment channels includes a pair of opposing channels formed along the vial.

4. The medication cartridge as set forth in claim 3, wherein the shell includes a pair of opposing slots that align with the pair of opposing channels.

5. The medication cartridge as set forth in claim 4, wherein the vial is slidable within the shell between an expanded state and a compressed state, such that when transitioning to the compressed state, the pump assembly is actuated to pump medicine from the outlet and through the dispensing aperture.

6. The medication cartridge as set forth in claim 5, wherein the medication cartridge has a bottom end and a top end, such that each of the pair of opposing slots has a first slot that rises upward from the bottom end toward the top end to a first junction, and then a second slot that traverses downward at an angle from the first junction toward a second junction, and then a third slot that rises upward again from the second junction toward the top end.

7. The medication cartridge as set forth in claim 6, wherein each of the pair of opposing channels rises upward from the bottom end toward the top end to the first junction, and then traverses downward at an angle from the first junction toward the second junction, where each of the pair of opposing channels terminates in a channel terminal.

8. The medication cartridge as set forth in claim 7, wherein the medication cartridge is formed such that depression of the top end of the medication cartridge forces the shell downward while the one or more alignment features pass through the third slots while remaining affixed within the channel terminals, thereby forcibly sliding the vial within the shell from the expanded state to the compressed state and actuating the pump assembly.

9. The medication cartridge as set forth in claim 2, wherein the vial is slidable within the shell between an expanded state and a compressed state, such that when transitioning to the compressed state, the pump assembly is actuated to pump medicine from the outlet and through the dispensing aperture.

10. The medication cartridge as set forth in claim 4, wherein the medication cartridge has a bottom end and a top end, such that each of the pair of opposing slots has a first slot that rises upward from the bottom end toward the top end to a first junction, and then a second slot that traverses downward at an angle from the first junction toward a second junction, and then a third slot that rises upward again from the second junction toward the top end.

11. The medication cartridge as set forth in claim 3, wherein each of the pair of opposing channels rises upward from the bottom end toward the top end to the first junction, and then traverses downward at an angle from the first junction toward the second junction, where each of the pair of opposing channels terminates in a channel terminal.

12. A medication cartridge, comprising: a shell having a dispensing aperture; a vial slidably positioned within the shell, the vial having a pump assembly with an outlet aligned with the dispensing aperture.

13. The medication cartridge as set forth in claim 12, wherein the vial includes a pair of opposing channels formed along the vial, the pair of opposing channels formed to align with and receive one or more alignment features of an oral medication dispensing device.

14. The medication cartridge as set forth in claim 13, wherein the shell includes a pair of opposing slots that align with the pair of opposing channels.

15. The medication cartridge as set forth in claim 14, wherein the vial is slidable within the shell between an expanded state and a compressed state, such that when transitioning to the compressed state, the pump assembly is actuated to pump medicine from the outlet and through the dispensing aperture.

16. The medication cartridge as set forth in claim 12, wherein the vial is slidable within the shell between an expanded state and a compressed state, such that when transitioning to the compressed

state, the pump assembly is actuated to pump medicine from the outlet and through the dispensing aperture.
